

VILNIUS, Lithuania

WMO: 267300

Lat: 54.6261N

Lon: 25.1069E

Elev: 156

StdP: 99.47

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.0	-22.2	0.5	-18.6	-19.0	0.7	-15.3	10.7	-2.0	9.9	-2.3	2.1	0	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.8	19.5	26.9	18.6	25.1	17.7	20.9	26.3	19.8	24.6	18.9	23.2	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	22.9	18.1	13.3	21.7	17.1	12.5	20.8	61.1	26.5	57.5	24.6	54.4	23.1	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.1	7.3	-22.9	31.7	3.6	1.6	-25.5	32.8	-27.6	33.8	-29.6	34.7	-32.2	35.8		
			-23.1	22.9	3.5	1.1	-25.6	23.7	-27.7	24.3	-29.7	24.9	-32.2	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-4.4	-3.5	0.5	7.5	12.7	16.1	18.5	17.5	12.6	6.8	1.9	-2.3
	DBStd	9.20	5.84	5.57	4.46	4.57	4.26	3.50	3.21	3.17	3.49	3.93	4.37	5.42
	HDD10.0	1994	446	377	295	101	21	1	0	0	14	114	243	382
	HDD18.3	4217	704	610	553	326	182	86	38	53	174	358	493	641
	CDD10.0	917	0	0	1	26	104	183	263	234	93	14	0	0
	CDD18.3	98	0	0	0	1	6	18	43	29	2	0	0	0
	CDH23.3	831	0	0	0	6	57	163	341	243	21	0	0	0
	CDH26.7	181	0	0	0	1	6	31	87	56	2	0	0	0
Wind	WSAvg	3.6	4.2	4.1	3.9	3.6	3.3	3.2	3.0	3.0	3.2	3.8	4.1	4.2
Precipitation	PrecAvg	692	48	43	42	42	61	66	94	75	62	58	49	52
	PrecMax	958	104	79	92	79	137	144	207	199	147	150	99	77
	PrecMin	504	10	18	14	7	29	17	13	17	12	7	10	21
	PrecStd	93	19	13	18	19	25	30	56	42	30	31	21	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.2	14.8	24.0	27.2	29.9	31.5	31.0	26.0	19.0	11.9	8.2
		MCWB	6.7	5.5	8.0	14.2	18.1	19.9	20.3	20.6	17.3	14.0	10.1	7.6
	2%	DB	4.8	5.2	11.7	20.9	25.0	27.1	29.1	28.0	22.9	16.0	10.1	6.8
		MCWB	4.4	4.0	6.6	12.4	16.7	18.5	19.8	18.8	16.5	13.0	8.9	6.1
	5%	DB	3.2	4.0	8.9	18.0	22.9	25.1	27.0	26.0	20.2	13.9	8.8	5.1
		MCWB	2.8	3.2	5.9	10.9	15.3	17.4	19.2	18.3	15.2	11.8	8.1	4.7
	10%	DB	2.1	2.9	6.6	15.8	20.8	23.1	25.0	24.0	18.3	12.2	7.2	3.8
		MCWB	1.8	2.3	4.4	9.6	14.2	16.3	18.4	17.5	14.2	10.7	6.8	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	6.1	9.5	15.2	19.3	21.1	22.8	22.1	19.0	15.1	10.7	8.0
		MCDB	7.1	6.9	12.9	22.9	24.9	27.1	28.3	27.8	23.8	17.5	11.5	8.1
	2%	WB	4.2	4.3	7.5	13.1	17.7	19.6	21.3	20.4	17.2	13.5	9.2	6.0
		MCDB	4.5	4.9	10.1	18.8	23.1	25.3	26.5	25.9	20.9	15.3	9.8	6.2
	5%	WB	2.9	3.1	6.0	11.7	16.3	18.4	20.2	19.4	16.0	12.2	7.9	4.6
		MCDB	3.1	3.6	8.2	16.7	21.2	23.2	24.6	24.0	19.2	13.6	8.5	5.0
	10%	WB	1.7	2.1	4.5	10.3	15.0	17.3	19.2	18.3	14.9	11.1	6.9	3.2
		MCDB	2.0	2.7	6.2	14.3	19.3	21.7	23.3	22.4	17.9	12.3	7.3	3.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	7.2	10.0	10.6	10.1	9.8	10.1	8.8	6.5	4.2	4.1
		MCDBR	4.1	5.1	10.5	13.8	13.3	13.0	12.5	12.9	11.6	8.8	5.5	4.2
		MCWBR	4.0	4.4	7.0	7.3	6.7	6.2	5.6	5.8	6.3	6.0	4.7	4.2
	5% WB	MCDBR	4.0	4.7	8.9	12.2	11.7	11.1	10.6	10.9	10.2	7.6	4.8	4.0
		MCWBR	4.0	4.3	6.6	7.1	6.7	6.0	5.6	5.7	6.1	5.7	4.4	4.1
Clear-Sky Solar Irradiance	taub	0.293	0.326	0.362	0.408	0.395	0.385	0.421	0.415	0.379	0.355	0.334	0.295	
	taud	2.186	2.153	2.193	2.190	2.300	2.355	2.272	2.283	2.356	2.398	2.281	2.218	
	Ebn at Noon	630	736	800	812	849	862	819	798	781	712	579	545	
	Edh at Noon	62	91	110	127	120	115	123	115	94	73	58	49	
All-Sky Solar Radiation	RadAvg	0.59	1.34	2.62	4.04	5.19	5.44	5.08	4.34	2.97	1.46	0.51	0.36	
	RadStd	0.11	0.25	0.42	0.51	0.52	0.48	0.52	0.40	0.31	0.30	0.07	0.07	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.44	N/A	N/A	N/A	+0.38	N/A	-126	-174	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.61	N/A	N/A	N/A

Nomenclature: See separate page